Sharing data as machine-actionable objects using RO-Crate, Bioschemas and Signposting

Stian Soiland-Reyes[1,2], Simone Leo[3], Leyla Jael Castro[4], Peter Sefton[5], Carole Goble[1]

[1] The University of Manchester, UK
[2] University of Amsterdam, Netherlands
[3] CRS4, Italy
[4] ZB Med, Germany
[5] The University of Queensland, Australia

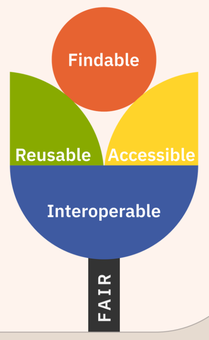
Practical and adaptable data sharing

Building a flexible FAIR infrastructure on the Web with **self-describing** digital objects handling several types of **metadata**

- Describe and package data collections, datasets, software etc. with their metadata

For real problems we also need…

- Platform-independent object exchange between repositories and services
- Representation of contextual elements
- Pragmatic best practices guides
- Domain-specific and general-purpose profiles
- Mixed object publication and archiving



Practical approaches

Infrastructure independent – avoiding repository/service silos *Practical, lightweight, robust*

Familiar, developer friendly, web native, machine and human readable, search engine accessible
Adoptable Linked Data JSON; guidance, not restrictions

Embrace diversity, legacy, unknowns, open-ended, multi-interpretation, self-describing, interlingua
Adaptable Metadata Profiles, add to existing Web resources

Bioschemas

https://bioschemas.org/

Bioschemas define domain-specific **profiles** to add structured metadata to Life Science resources on the Web by using and expanding **schema.org**: *ChemicalSubstance, Gene, MolecularEntity, Protein, ProteinStructure, Sample, Taxon, …*

General-purpose profiles are being lifted for use beyond life sciences (https://schemas.science/) including *Dataset, Course, ComputationalWorkflow, ComputationalTool, TrainingMaterial.*

Bioschemas are **deployed** across **>67 ELIXIR resources** and beyond, covering **>180** profile deployments overall.

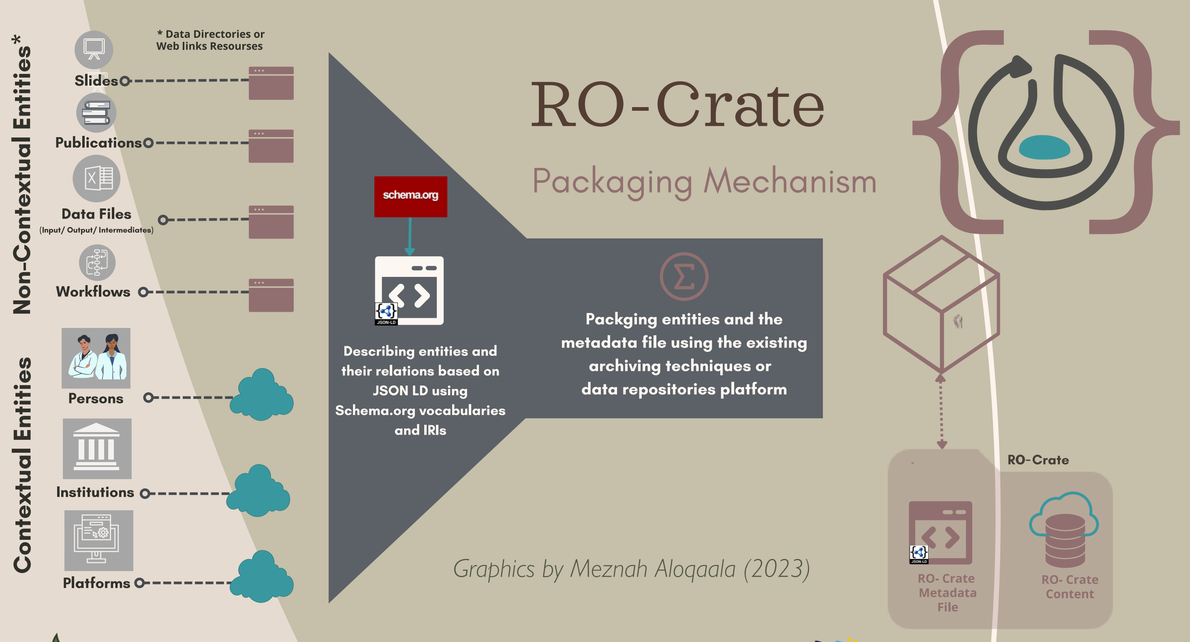
Bioschemas are developed and maintained by working groups in an active **community** (>150 members)

Signposting

FAIR Signposting uses standard HTTP `Link` headers (RFC8288) for any resource type (including HTML landing pages). *Machine agents* then use HTTP `HEAD` to predictably find the individual components of a FAIR resource: *PID, authors, licence, metadata* and *dataset download*.

Any HTTP resource can become full FDOs by providing their own Signposting, with no other changes to their infrastructure.

https://signposting.org/FAIR/ https://pypi.org/project/signposting/



RO-Crate in practice

Computational Workflows
Biosciences, Climate science, Biodiversity
EOSC-Life Research Infrastructure Cluster: The computational workflow registry and its services import, export, store and publish RO-Crates, to support the full workflow life cycle. Technology is domain-agnostic and adopted outside bioscience.

Data Cubes – *tabular data, Earth Science, Bioscience*
The EOSC project *RELIANCE* use RO-Crate to package data cubes of **earth observation data**, along with documentation, images and related infrastructures. Metadata includes **temporal coverage, spatial coverage and vertical coverage**. ROHub publishes the archived RO-Crates to general-purpose repositories (Zenodo, B2Share) for longevity and PIDs.

Mixed Object publishing and repository exchange
The *Helmholtz HMC Hub Energy* uses RO-Crates to move time series data from different databases exported with metadata description of their structure and content into a single web service. The HERMES project uses RO-Crates for software publication pipelines

Repository exchange and archiving
language studies and cultural heritage
PARADISEC and the *Language Data Commons of Australia* use RO-Crate for language data large text corpuses with **personally-identifiable information**. Adds **granular access control** and **restriction of use** on individual texts within the larger RO-Crate, which metadata can be open.
Jupyter Notebooks give programmatical access to crate content for analytics, selecting text by general and domain-specific metadata.

Executing Data and Software Management Plans
RO-Crates are combined with machine-actionable **Data Management Plans** (maDMPs) and Software Management Plans to automate and facilitate management of research data. Within *ELIXIR*, RO-Crate will integrate the **Data Stewardship Wizard** with Galaxy workflows to automate FDO creation that also follows data management plans.

RO-Crate and Bioschemas at ELIXIR AHM2023
Workshop 10: *Building lightweight FAIR data packages with Bioschemas and RO-Crate* (Tue 11:00)
Workshop 3: *For sustainable Agriculture: Status, Collaborations and Perspectives of the Plant Community within ELIXIR* (Tue 09:00)
Mini-symposium 6: *Reproducible Analytics & Infrastructure*
Workshop 17: *Embedding a new data resource in the ELIXIR ecosystem – a practical guide* (Wed 11:00)
Poster 11: *de.NBI-GCBN - The German contribution to the Plant Community of ELIXIR*
Poster 53: *From launch to maturity: growing and sustaining the RDMkit*
Poster 87: *Enabling an operational, open and FAIR EOSC ecosystem*
Poster 89: *WorkflowHub – a registry for describing, sharing and publishing scientific computational workflows*
Poster 92: *Improving Reproducibility of Computational Analyses Performed on the ELIXIR-GR Cloud*
Poster 93: *Making workflow provenance FAIR across workflow systems with Workflow Run RO-Crate*

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https://bioschemas.org/
https://researchobject.org/ro-crate
https://doi.org/10.5281/zenodo.7984529

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